Week 11 Assignment

Logistic regression and other GLMs

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 12′s lecture (Monday). See the syllabus for the late policy.

Part 1: Conceptual

For each outcome, name the outcome type and the model you'd use (`lm`, logistic `glm`, or Poisson/negative-binomial `glm`):

1. Clutch size (number of eggs laid)
2. Whether a plant flowers this season (yes/no)
3. Leaf area in cm²
4. Number of ticks found on an individual host

Part 2: Applied

Using this week's tutorial's simulated `seedlings` data (or re-simulate it with a different `set.seed()`):

1. Fit `glm(survived ~ light, family = binomial)`.
2. Exponentiate the `light` coefficient and interpret it as an odds ratio, in a sentence a non-statistician biologist would understand.
3. Using the simulated `counts` data, fit the Poisson model and check for overdispersion (residual deviance vs. residual df).

Is overdispersion a concern here?